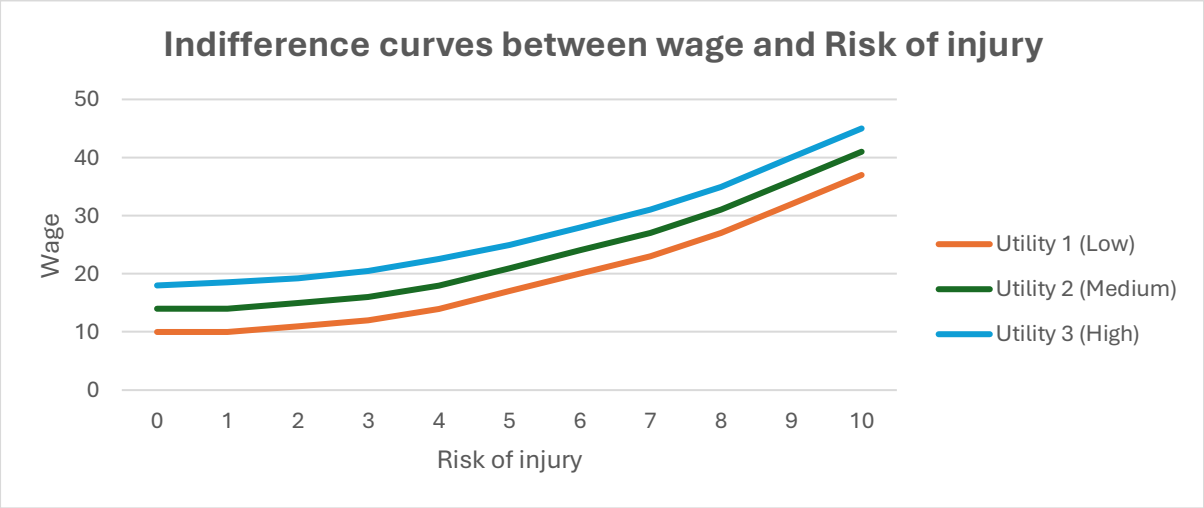
Labour economics coursework 2

a) Adam Smith's compensating wage differentials with a sustainability discount.

b) Adam Smith's idea is that job characteristics influence labour market equilibrium. In the typical supply-demand model, the wage allocates workers across firms until the wage equals the value of the marginal product. In the theory of compensating wage differentials, the utility function that defines a worker's preferences depends not only on wages but also on the amenities provided by the employer. A typical example in this theory is the risk of injury the worker faces on the job. In the example of the Market for Risky Jobs, the utility would be represented like this:

$$U = f(w, prob\ of\ injury)$$

In this case, the Marginal utility of income would be positive. The higher the probability of injury, the higher the salary; thus, the likelihood of injury reduces utility, which the wage must compensate. This example of a market where you consider the risk of injury could be illustrated like this:

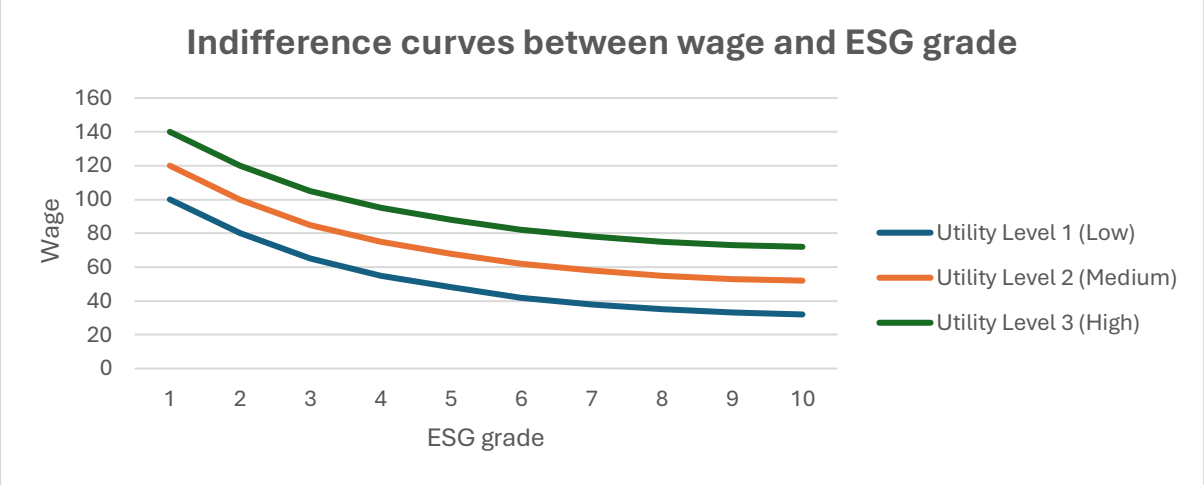


According to Atz et al. (2023), the paper begins by noting a correlation between a company's performance and its ESG rating. Kruger et al. (2024) aim to explain the link between financial performance and sustainability. According to them, companies with a good ESG profile often benefit from strong FCFs (financial cash flow). The theory they present is that this financial cash flow is enabled by a discounted labour force. In the same way, risky jobs have to pay workers more to compensate for the risk and to equate the utility they get from this job with that of a non-risky job. Firms that are considered sustainable can pay people less because they have a positive impact on society, thereby increasing their workers' utility. Overall, the ESG score is considered an amenity that increases utility, so the monetary wage must fall to maintain an equilibrium, creating a "sustainability discount" where w, sustainable < w, unsustainable. The difference from the example of risky jobs is that here the amenities are non-pecuniary.

The conclusion of the Kruger et al. (2024) study found that the sustainability wage gap is 8.7% at the baseline. If we assume that this conclusion falls under the same theory as the risky job market, we could illustrate it like this:

$$U = f(w, ESG)$$

Where w represents wage and (ESG) is the ESG grade that has a positive impact on the utility of the worker. Here we have a graph showing the relationship between ESG grade and wage, assuming a negative correlation.



To calculate those numbers, the paper used employer-employee matched data from Sweden. To avoid omitted-variable bias related to human capital, they included variables such as scores on cognitive tests administered by the army and years of experience. The use of panel data, which allowed them to observe workers changing jobs and classify the reasons for those changes, enabled them to obtain precise results. This is important because a worker's innate ability does not vary across jobs, so establishing the correlation between wage changes and job amenities is a better measure of wage differentials.

With all that information and Adam Smith's theory of wage compensation, we can say that workers with strong environmental preferences are willing to forgo part of their wage for meaningful social impact, which we could treat as a non-pecuniary amenity, aligning with Adam Smith's theory.

c) We want to assess the impact of wage sustainability; we will use the log wage as the dependent variable. The main explanatory variable we are going to examine is the dummy variable sustainable.

To avoid omitted-variable bias, we will consider other explanatory variables within the human capital theory. Our goal is to include all potential explanatory variables that could bias our analysis of the influence of sustainability on wages. We will include the following variables: (**EDUC**), a very important variable that could affect the choice to work in a sustainable company, and the wage. (**EXPER)** and (**EXPERSQ)** experience can influence wages, and experience squared captures diminishing returns to experience. (**BAS3)** To take into account cognitive capacity that could affect both the choice to work in a sustainable firm and the wage, and (**KWW)** to take into account working knowledge. Finally, we will include dummy variables for **(RACE), (FEMALE)**, and demographics to ensure that the observed gap is not driven by demographic disparities or by differences between **(RURAL)** and urban areas, which can significantly affect wages.

The regression formula is the following:

$$lwageit = \beta 0 + \beta 1 SUSTAINABLEit + \beta 2 EDUCit + \beta 3 EXPERit + \beta 4 EXPERit2 + \beta 5 FEMALEit + \beta 6 RACEit + \beta 7 RURALit + \beta 8 KWW + \beta 9 BAS3it + uit$$

According to the theory of wage compensation, the beta coefficient should be negative because workers are willing to accept lower wages when working for a sustainable company. The hypotheses are as follows:

H1 (Alternative): β1<0 and is statistically significant (The Sustainability Discount exists).

H0 (Null): β1≥0 or is not statistically significant.

The calculation and researcher will only support the paper if the first hypothesis is verified.

d)

Table 1: OLS Estimates of the Sustainability Wage Gap

	(1) Log Hourly Wage
Sustainable Job	-0.049 (0.030)
Years of Schooling	0.060*** (0.006)
Experience	0.082*** (0.010)
Experience Squared	-0.002*** (0.000)
Intelligence Score	0.002** (0.001)
Female	-0.027 (0.017)
Non-White	-0.084*** (0.028)
Rural Area	-0.060** (0.030)
Work Knowledge Score	0.007*** (0.001)
Constant	4.590*** (0.116)
Observations	2033
R^2	0.212

Standard errors in parentheses
Standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Overall, we see that many of our variables affect wages; most of the variables we included to account for human capital also affect wages. The following table describes the impact of each of those variables on the variable **(LogWage):**

Variable	Type	Coefficient	Impact on Hourly Wage (%)	Interpretation
Race (non white)	Dummy	-0.084	-8.06%	Nonwhite person earn than white workers
Experience	Continuous	0.082	+8.20%	Every year of experience has a positive impact on wage
Education	Continuous	0.060	+6.00%	Each year of schooling increase wage
Rural Area	Dummy	-0.060	-6.00%	Rural areas earn less than urban areas
KWW Score	Continuous	0.007	+0.70%	Knowledge of work world have a positive impact on wage
BAS3 (Intelligence)	Continuous	0.002	+0.20%	Intelligence has a positive impact on wage

Regarding our main variable of interest, we find that sustainable workers earn 4.9% less than non-sustainable workers. However, this variable is statistically insignificant, so the result only partially supports the theory. The empirical results are not satisfying, so we have to reject the hypothesis that beta is negative. In the meantime, we can still say that the fact that are variable sustainable is non significant does not make the result of the Krueger et al. (2024) paper invalide, the difference in methodology and the use of a cross sectional data with a dummy variable representing the sustainibility of the firm is, as we can say not sufficient to support the papers finding but does reject the establishement of a link between the Compensating wage differential and the existence of a "sustainable discount".

e) A limitation of the database is that it includes only a dummy variable for sustainability, which limits the interpretation of the impact of ESG grades on workers' wages. In contrast, the research paper uses ESG grades, which makes the results on salary evolution much more interesting and opens up many other avenues for research. Even if the paper uses an ESG rating instead of a dummy variable for sustainability, there is also a question about the limitations of ESG ratings and their methodologies, and the authors admitted that the ESG rating methodology is a black box with low visibility.

Another limitation of our analysis stems from the nature of our data. We use a cross-sectional dataset, which limits our ability to follow individuals over the evolution of their careers. The paper by Kruger et al. (2024) uses a longitudinal panel dataset that enables them to track individuals over the evolution of their careers. This allows them to use individual fixed effects to eliminate all time-invariant unobserved heterogeneity. According to George J. Bordjas (Labour Economics), *"Because a worker's innate ability does not change from job to job, the correlation between the change in wage and in*

amenity might better measure the compensation of wage differential". This implies that an analysis limited to a simple observation affixes a moment in time as a significant limitation to establishing correlation.

The paper relates the switch between sustainable and unsustainable sectors by placing it in the context of exogenous events that could have a significant impact on those decisions, using richer data on work variation. This is a major limitation in our analysis because this unobserved exogenous event could bias our calculation.

The last limitation common to the estimation we made and the paper is the absence of information about other pecuniary amenities across different sectors of work.

After every limitation we pointed out, the dataset used in the research paper by Kruger et al. (2024) is more suitable for answering the question about the mechanism that makes sustainable companies more competitive, why they have lower labour costs, and where they come from. The two main reasons are the nature of the data, which can eliminate certain exogenous events by observing the evolution over time. The second reason their data is more suitable is the use of ESG ratings, even though they present certain limitations.

f)

```
*First thing we need to do is open the data set
use "C:\Users\adgf918\OneDrive - City, University of London\Desktop\Labor economics coursework 2\MainData"
*Do file for the EC3019 Labour economics Coursework 2
describe
summarize
*Those two commands are used to be able to take a better look at the data to start understanding what the variable of interest is and in what state they are they
* The first observation we can make is that we have to clean the data, drop certain variables, and convert other variables into numbers, like gender
*The first modification we have to do is to desrting data
destring educ age exper, replace
*Since I destring those variables, I can now take a look at them with the following command
summarise educ age exper
*We can also see that the variable gender is under the following form: male, female, and we want to transform it into a dummy variable.
gen female_dummy = (gender=="female")
tab female_dummy
*This gives us a new variable that describes the gender as 1=female and 0=male, which makes it usable for our regressions
*We also have the variable enrol, which is a string, so we are going to destring it
destring enrol, replace
*We now have to standardise the variable unsus into a dummy variable
gen sustainable = (unsus=="sustainable")
tab sustainable
*We now have unsustainable, which is equal to 0, and sustainable, which is equal to 1
*Question c) and d)
*Now that we have chosen all of the variables, we are just missing one, which is experience squared, that we are going to generate
gen expersq = exper*exper
*------------------------------
*We can now run the regression
regress lwage i.sustainable educ exper expersq BAS3 i.female_dummy i.race i.rural KWW, robust
estimates store model1
ssc install estout
*Generating the table
esttab model1 using "ResultsTable.rtf", replace ///
  b(3) se(3) r2(3) ///
  star(* 0.10 ** 0.05 *** 0.01) ///
  nobaselevels ///
  label ///
  title("Table 1: OLS Estimates of the Sustainability Wage Gap") ///
  mtitles("Log Hourly Wage") ///
  coeflabels( ///
    1.sustainable "Sustainable Job" ///
    educ "Years of Schooling" ///
    exper "Experience" ///
    expersq "Experience Squared" ///
    BAS3 "Intelligence Score" ///
    1.female_dummy "Female" ///
    1.race "Non-White" ///
    1.rural "Rural Area" ///
    KWW "Work Knowledge Score" ///
    _cons "Constant" ///
  ) ///
  addnotes("Standard errors in parentheses")
```